FoodRescue

DBMS Mini Project — SQL DDL commands executions

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Creating the Database:

```
(PES2UG24CS200_PES2UG24CS228_29@root) [(none)]>create Database FoodRescue;
Query OK, 1 row affected (0.02 sec)
```

Making the following tables according to our schema and problem statement:

#	Table Name	Primary Key	Attributes
1	**DONOR**	donor_id	Name, Donor_Type, Contact_Person, Phone_No, Email, Street, City, State, Country
2	**RECIPIENT_ORGANIZATION**	recipient_id	Organization_Name, Organization_Type, Contact_Person, Phone_No, Email, Street, City, State, Country
3	**FOOD_BATCH**	batch_id	donor_id (FK), Food_Name, Food_Category, Quantity, Unit, Condition_Status, Production_Date, Expiry_Date, Batch_Status
4	**COLLECTION_REQUEST**	request_id	batch_id (FK), recipient_id (FK), Requested_Quantity, Request_Date, Preferred_Pickup_Time, Request_Status, Notes
5	**TRANSPORTATION**	transport_id	request_id (FK), Driver_Name, Vehicle_Number, Pickup_Time, Delivery_Time, Transport_Status
6	**DISTRIBUTION**	distribution_id	request_id (FK), recipient_id (FK), Distributed_Quantity, Distribution_Date, Distribution_Status, Received_By

Creating donor table:

```
(PES2UG24CS200_PES2UG24CS228_29@root) [foodrescue]>CREATE TABLE DONOR (
    ->     donor_id         INT AUTO_INCREMENT PRIMARY KEY,
    ->     Name             VARCHAR(100) NOT NULL,
    ->     Donor_Type       VARCHAR(50),
    ->     Contact_Person   VARCHAR(100),
    ->     Phone_No         VARCHAR(15),
    ->     Email            VARCHAR(100),
    ->     Street           VARCHAR(100),
    ->     City             VARCHAR(50),
    ->     State            VARCHAR(50),
    ->     Country          VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.04 sec)
```

Verifying if donor table created correctly:

```
(PES2UG24CS200_PES2UG24CS228_29@root) [foodrescue]>desc DONOR;
+----------------+--------------+------+-----+---------+----------------+
| Field          | Type         | Null | Key | Default | Extra          |
+----------------+--------------+------+-----+---------+----------------+
| donor_id       | int          | NO   | PRI | NULL    | auto_increment |
| Name           | varchar(100) | NO   |     | NULL    |                |
| Donor_Type     | varchar(50)  | YES  |     | NULL    |                |
| Contact_Person | varchar(100) | YES  |     | NULL    |                |
| Phone_No       | varchar(15)  | YES  |     | NULL    |                |
| Email          | varchar(100) | YES  |     | NULL    |                |
| Street         | varchar(100) | YES  |     | NULL    |                |
| City           | varchar(50)  | YES  |     | NULL    |                |
| State          | varchar(50)  | YES  |     | NULL    |                |
| Country        | varchar(50)  | YES  |     | NULL    |                |
+----------------+--------------+------+-----+---------+----------------+
10 rows in set (0.01 sec)
```

Creating recipient organization table

```
(PES2UG24CS200_PES2UG24CS228_29@root) [foodrescue]>CREATE TABLE RECIPIENT_ORGANIZATION (
    ->     recipient_id       INT AUTO_INCREMENT PRIMARY KEY,
    ->     Organization_Name  VARCHAR(100) NOT NULL,
    ->     Organization_Type  VARCHAR(50),
    ->     Contact_Person     VARCHAR(100),
    ->     Phone_No           VARCHAR(15),
    ->     Email              VARCHAR(100),
    ->     Street             VARCHAR(100),
    ->     City               VARCHAR(50),
    ->     State              VARCHAR(50),
    ->     Country            VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

Verifying if recipient organization table is made properly:

```
(PES2UG24CS200_PES2UG24CS228_29@root) [foodrescue]>desc Recipient_Organization;
+-------------------+--------------+------+-----+---------+----------------+
| Field             | Type         | Null | Key | Default | Extra          |
+-------------------+--------------+------+-----+---------+----------------+
| recipient_id      | int          | NO   | PRI | NULL    | auto_increment |
| Organization_Name | varchar(100) | NO   |     | NULL    |                |
| Organization_Type | varchar(50)  | YES  |     | NULL    |                |
| Contact_Person    | varchar(100) | YES  |     | NULL    |                |
| Phone_No          | varchar(15)  | YES  |     | NULL    |                |
| Email             | varchar(100) | YES  |     | NULL    |                |
| Street            | varchar(100) | YES  |     | NULL    |                |
| City              | varchar(50)  | YES  |     | NULL    |                |
| State             | varchar(50)  | YES  |     | NULL    |                |
| Country           | varchar(50)  | YES  |     | NULL    |                |
+-------------------+--------------+------+-----+---------+----------------+
10 rows in set (0.00 sec)
```

Creating Food Batch table:

```
(PES2UG24CS200_PES2UG24CS228_29@root) [foodrescue]>CREATE TABLE FOOD_BATCH (
    ->     batch_id          INT AUTO_INCREMENT PRIMARY KEY,
    ->     donor_id          INT NOT NULL,
    ->     Food_Name         VARCHAR(100) NOT NULL,
    ->     Food_Category     VARCHAR(50),
    ->     Quantity          DECIMAL(10,2) NOT NULL,
    ->     Unit              VARCHAR(20),
    ->     Condition_Status  VARCHAR(50),
    ->     Production_Date   DATE,
    ->     Expiry_Date       DATE NOT NULL,
    ->     Batch_Status      VARCHAR(30) DEFAULT 'Available',
    ->     CONSTRAINT fk_foodbatch_donor
    ->         FOREIGN KEY (donor_id) REFERENCES DONOR(donor_id)
    ->         ON DELETE RESTRICT ON UPDATE CASCADE,
    ->     CONSTRAINT chk_batch_status
    ->         CHECK (Batch_Status IN ('Available','Reserved','Collected','Expired')),
    ->     CONSTRAINT chk_quantity_positive
    ->         CHECK (Quantity > 0)
    -> );
Query OK, 0 rows affected (0.03 sec)
```

Verifying if FoodBatch table made properly:

```
(PES2UG24CS200_PES2UG24CS228_29@root) [foodrescue]>desc FOOD_BATCH;
+------------------+---------------+------+-----+-----------+----------------+
| Field            | Type          | Null | Key | Default   | Extra          |
+------------------+---------------+------+-----+-----------+----------------+
| batch_id         | int           | NO   | PRI | NULL      | auto_increment |
| donor_id         | int           | NO   | MUL | NULL      |                |
| Food_Name        | varchar(100)  | NO   |     | NULL      |                |
| Food_Category    | varchar(50)   | YES  |     | NULL      |                |
| Quantity         | decimal(10,2) | NO   |     | NULL      |                |
| Unit             | varchar(20)   | YES  |     | NULL      |                |
| Condition_Status | varchar(50)   | YES  |     | NULL      |                |
| Production_Date  | date          | YES  |     | NULL      |                |
| Expiry_Date      | date          | NO   |     | NULL      |                |
| Batch_Status     | varchar(30)   | YES  |     | Available |                |
+------------------+---------------+------+-----+-----------+----------------+
10 rows in set (0.00 sec)
```

Creating a collection_request table:

```
(PES2UG24CS200_PES2UG24CS228_29@root) [foodrescue]>CREATE TABLE COLLECTION_REQUEST (
    ->     request_id             INT AUTO_INCREMENT PRIMARY KEY,
    ->     batch_id               INT NOT NULL,
    ->     recipient_id           INT NOT NULL,
    ->     Requested_Quantity     DECIMAL(10,2) NOT NULL,
    ->     Request_Date           DATE NOT NULL,
    ->     Preferred_Pickup_Time  DATETIME,
    ->     Request_Status         VARCHAR(30) DEFAULT 'Pending',
    ->     Notes                  VARCHAR(255),
    ->     CONSTRAINT fk_request_batch
    ->         FOREIGN KEY (batch_id) REFERENCES FOOD_BATCH(batch_id)
    ->         ON DELETE RESTRICT ON UPDATE CASCADE,
    ->     CONSTRAINT fk_request_recipient
    ->         FOREIGN KEY (recipient_id) REFERENCES RECIPIENT_ORGANIZATION(recipient_id)
    ->         ON DELETE RESTRICT ON UPDATE CASCADE,
    ->     CONSTRAINT chk_request_status
    ->         CHECK (Request_Status IN ('Pending','Approved','Rejected','Completed')),
    ->     CONSTRAINT chk_requested_qty_positive
    ->         CHECK (Requested_Quantity > 0)
    -> );
Query OK, 0 rows affected (0.03 sec)
```

Verifying If collection request is made correctly:

```
(PES2UG24CS200_PES2UG24CS228_29@root) [foodrescue]>desc collection_request;
+----------------------+---------------+------+-----+---------+----------------+
| Field                | Type          | Null | Key | Default | Extra          |
+----------------------+---------------+------+-----+---------+----------------+
| request_id           | int           | NO   | PRI | NULL    | auto_increment |
| batch_id             | int           | NO   | MUL | NULL    |                |
| recipient_id         | int           | NO   | MUL | NULL    |                |
| Requested_Quantity   | decimal(10,2) | NO   |     | NULL    |                |
| Request_Date         | date          | NO   |     | NULL    |                |
| Preferred_Pickup_Time | datetime     | YES  |     | NULL    |                |
| Request_Status       | varchar(30)   | YES  |     | Pending |                |
| Notes                | varchar(255)  | YES  |     | NULL    |                |
+----------------------+---------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)
```

Creating the transportation table:

```
(PES2UG24CS200_PES2UG24CS228_29@root) [foodrescue]>CREATE TABLE TRANSPORTATION (
    ->     transport_id      INT AUTO_INCREMENT PRIMARY KEY,
    ->     request_id        INT NOT NULL,
    ->     Driver_Name       VARCHAR(100),
    ->     Vehicle_Number    VARCHAR(20),
    ->     Pickup_Time       DATETIME,
    ->     Delivery_Time     DATETIME,
    ->     Transport_Status  VARCHAR(30) DEFAULT 'Scheduled',
    ->     CONSTRAINT fk_transport_request
    ->         FOREIGN KEY (request_id) REFERENCES COLLECTION_REQUEST(request_id)
    ->         ON DELETE RESTRICT ON UPDATE CASCADE,
    ->     CONSTRAINT chk_transport_status
    ->         CHECK (Transport_Status IN ('Scheduled','In_Transit','Delivered','Cancelled'))
    -> );
Query OK, 0 rows affected (0.03 sec)
```

Verifying transportation table is created correctly

```
(PES2UG24CS200_PES2UG24CS228_29@root) [foodrescue]>desc Transportation;
+------------------+--------------+------+-----+------------+----------------+
| Field            | Type         | Null | Key | Default    | Extra          |
+------------------+--------------+------+-----+------------+----------------+
| transport_id     | int          | NO   | PRI | NULL       | auto_increment |
| request_id       | int          | NO   | MUL | NULL       |                |
| Driver_Name      | varchar(100) | YES  |     | NULL       |                |
| Vehicle_Number   | varchar(20)  | YES  |     | NULL       |                |
| Pickup_Time      | datetime     | YES  |     | NULL       |                |
| Delivery_Time    | datetime     | YES  |     | NULL       |                |
| Transport_Status | varchar(30)  | YES  |     | Scheduled  |                |
+------------------+--------------+------+-----+------------+----------------+
7 rows in set (0.00 sec)
```

Creating distribution table:

```
(PES2UG24CS200_PES2UG24CS228_29@root) [foodrescue]>CREATE TABLE DISTRIBUTION (
    ->     distribution_id       INT AUTO_INCREMENT PRIMARY KEY,
    ->     request_id            INT NOT NULL,
    ->     recipient_id          INT NOT NULL,
    ->     Distributed_Quantity  DECIMAL(10,2) NOT NULL,
    ->     Distribution_Date     DATE NOT NULL,
    ->     Distribution_Status   VARCHAR(30) DEFAULT 'Completed',
    ->     Received_By           VARCHAR(100),
    ->     CONSTRAINT fk_distribution_request
    ->         FOREIGN KEY (request_id) REFERENCES COLLECTION_REQUEST(request_id)
    ->         ON DELETE RESTRICT ON UPDATE CASCADE,
    ->     CONSTRAINT fk_distribution_recipient
    ->         FOREIGN KEY (recipient_id) REFERENCES RECIPIENT_ORGANIZATION(recipient_id)
    ->         ON DELETE RESTRICT ON UPDATE CASCADE,
    ->     CONSTRAINT chk_distribution_status
    ->         CHECK (Distribution_Status IN ('Pending','Completed','Partially_Completed')),
    ->     CONSTRAINT chk_distributed_qty_positive
    ->         CHECK (Distributed_Quantity > 0)
    -> );
Query OK, 0 rows affected (0.04 sec)
```

Verifying if distribution table is correctly made

```
(PES2UG24CS200_PES2UG24CS228_29@root) [foodrescue]>desc distribution
    -> ;
+----------------------+---------------+------+-----+-----------+----------------+
| Field                | Type          | Null | Key | Default   | Extra          |
+----------------------+---------------+------+-----+-----------+----------------+
| distribution_id      | int           | NO   | PRI | NULL      | auto_increment |
| request_id           | int           | NO   | MUL | NULL      |                |
| recipient_id         | int           | NO   | MUL | NULL      |                |
| Distributed_Quantity | decimal(10,2) | NO   |     | NULL      |                |
| Distribution_Date    | date          | NO   |     | NULL      |                |
| Distribution_Status  | varchar(30)   | YES  |     | Completed |                |
| Received_By          | varchar(100)  | YES  |     | NULL      |                |
+----------------------+---------------+------+-----+-----------+----------------+
7 rows in set (0.00 sec)
```